

Mofe Omolade

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EDUCATION

University of British Columbia, *BASc in Computer Engineering*

Expected Graduation May 2029

Relevant Coursework: Circuit Analysis, Electricity & Magnetism, Linear Algebra, Embedded C, Engineering Design.

WORK EXPERIENCE

FulcrumAir, *EE Intern*, Calgary, AB

May 2026 - Present

- Designed and integrated a custom motor driver PCB around the IFX007T, MAX33040E, and LMR50410-Q1.
- Wrote C++ firmware for multichannel motor control, CAN telemetry transmission, and actuator feedback calculation.
- Revised past schematics and PCB layout to implement electrical best practices and improve electronic reliability.

University of Calgary, *EE Research Assistant*, Calgary, AB

July 2023 - August 2023

- Modelled and simulated Op-Amp oscillator circuits in LTspice to support research in amplifier-based oscillators.
- Contributed to academic projects across faculties and presented findings at symposiums and research boards.

PROJECTS

Speedometer Heads-Up Display

- Developed a low-latency heads-up display using an Arduino Nano, GY-NEO6MV2 module, and SPI OLED display.
- Wrote C++ firmware to parse NMEA data streams to compute velocity within 5% of the factory speedometer.
- Integrated power regulation circuit to step down 14V vehicle outputs to a stable 5V rail, to protect logic circuitry.

Autonomous 6-Wheel Rover

- Designed power and drivetrain prototype around the L298N full-bridge motor driver and Nucleo F446RE MCU.
- Implemented C firmware for command parsing, Hall Effect sensor feedback, and UART serial communication.

SKILLS

- *Languages:* C/C++, Python
- *Tools:* Altium, KiCAD, MATLAB, LTspice, VS Code, Git, Microsoft Excel.
- *Hardware & Lab:* Arduino, STM32, ESP32, Oscilloscopes, Soldering, Circuit Debugging.

LEADERSHIP EXPERIENCE

Place Vanier Residence Association, *VP Finance*, Vancouver, BC

September 2025 - Present

- Led the events subcommittee to plan and execute initiatives, managing timelines, budgets, and communications.
- Leveraged resident feedback to tailor events, significantly boosting attendance at house events by over 100%.

Society for AI Literacy, *Outreach Director Executive*, Vancouver, BC

October 2023 - Present

- Organized and executed AI workshops for 300+ students, managing outreach, curriculum, and event coordination.
- Secured partnerships with academic and industry organizations to expand educational programming.